

## Class #13 - Characterization of an invertible matrix

1. Each elementary matrix is invertible:

Recall: every row operations are \_\_\_\_\_.

If  $E$  is produced by a row operation on  $I$ , then there is  $F$  that changes  $E$  back into  $I$ . Since  $E$  and  $F$  correspond to reverse operations, \_\_\_\_\_  $= I =$  \_\_\_\_\_.

2. Example: Find the inverse of  $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$

3. Theorem: An  $n \times n$  matrix  $A$  is invertible if and only if  $A$  is row equivalent to \_\_\_\_\_.

Proof: Suppose  $A$  is invertible. Then  $Ax = b$  is \_\_\_\_\_ for \_\_\_\_\_  $b \in \mathbf{R}^n$ . Therefore, there is a \_\_\_\_\_ in every \_\_\_\_\_ in its echelon form. Since  $A \in \mathbf{R}^{n \times n}$ , \_\_\_\_\_.

Now suppose  $A$  is row equivalent to  $I$ . Then there are elementary matrices  $E_1, \dots, E_k$  such that

$$E_k E_{k-1} \cdots E_3 E_2 E_1 A = \text{_____}$$

Since each  $E_i$  is invertible, we get

Hence,  $A$  is \_\_\_\_\_ and  $A^{-1} =$  \_\_\_\_\_.

Note: Sequence of row operations that reduces  $A$  to  $I$  also transforms  $I$  to \_\_\_\_\_.

4. Algorithm for finding  $A^{-1}$ :

Apply row operations to the matrix  $[A \quad I]$  to reduce the matrix  $A$  into  $I$ . It will result in  $[I \quad A^{-1}]$ . If  $A$  can't be reduced to  $I$ , then  $A$  is \_\_\_\_\_.

Example: Find the inverse of  $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ , if it exists.

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5. Question: Suppose  $CA = I$ . Show that the equation  $Ax = 0$  has only the trivial solution.

6. Question: Suppose  $AD = I$ . Show that  $Ax = b$  is consistent for all  $b \in \mathbf{R}^n$ .

7. The Invertible Matrix Theorem: Let  $A$  be a square  $n \times n$  matrix. TFAE:

- (a)  $A$  is an invertible matrix.
- (b)  $A$  is row equivalent to \_\_\_\_\_.
- (c)  $A$  has \_\_\_\_\_pivot positions.
- (d) The equation  $Ax = 0$  has \_\_\_\_\_solution.
- (e) The columns of  $A$  form a \_\_\_\_\_set.
- (f) The linear transformation  $T(x) = Ax$  is \_\_\_\_\_.
- (g) The equation  $Ax = b$  is \_\_\_\_\_for \_\_\_\_\_ $b \in$  \_\_\_\_\_.
- (h) The columns of  $A$  span \_\_\_\_\_
- (i) The linear transformation  $T(x) = Ax$  is \_\_\_\_\_ $\mathbf{R}^n$ .
- (j) There is an  $n \times n$  matrix  $C$  such that  $CA = I$ .
- (k) There is an  $n \times n$  matrix  $D$  such that  $AD = I$ .
- (l)  $A^T$  is an \_\_\_\_\_matrix.

8. Note: This tells us that

- (a) if  $A$  and  $C$  are both  $n \times n$  square matrices with  $AC = I$ , then  $A$  is \_\_\_\_\_. Moreover,  $A^{-1} =$  \_\_\_\_\_.

(don't have check both sides in the inverse definition for square matrices)

- (b) if  $A$  and  $B$  are  $n \times n$  matrices and  $AB = I$ , then \_\_\_\_\_ is invertible, and \_\_\_\_\_ is invertible. Moreover,  $B =$  \_\_\_\_\_ and  $A =$  \_\_\_\_\_.